

Ting-Yun (Charlotte) Chang

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RESEARCH INTERESTS

Inference Efficiency Reducing LLM serving costs through model quantization and KV cache compression
Actionable Interp. Diagnosing model behavior and translating scientific insights into actionable methods
Multimodality. Improving unified generative models that integrate vision and language

EDUCATION

University of Southern California, USA 2021 - Spring 2026
PhD student in Department of Computer Science
(All but dissertation)

National Taiwan University, Taiwan 2018 - 2020
M.S. in Department of Computer Science and Information Engineering

National Tsing Hua University, Taiwan 2014 - 2018
B.S. in Department of Computer Science GPA 4.14/4.3

Tsinghua University, China Fall 2015
Exchange Student in Department of Computer Science and Technology

RESEARCH EXPERIENCE

University of Southern California California, USA
Research Assistant 2021 - Present

- Advisors: Robin Jia and Jesse Thomason
- KV cache compression for reasoning models with sparse attention and quantization methods
- Study why low-bit model quantization affects certain examples disproportionately [1]
- Improve LLMs' consistency to prompt variants by model decomposition and efficient weight updates [3]
- Localize where memorized data resides in LLM parameters and unlearn them [4]
- Improve the stability of in-context learning to prompt choices via data valuation of demonstrations [5]
- Benchmark continual learning abilities of vision-language models with a suite of multimodal tasks [6]

NVIDIA California, USA
Gen-AI SWE Intern Spring 2026

- Post-train Cosmos video diffusion models and develop benchmark evaluation

Google DeepMind New York, USA
Research Intern Summer 2025

- Hosts: Xiang Zou, Zun Li, Jiao Sun, and Shyam Upadhyay
- Investigate the interplay between RL algorithms and post-training model quantization

Amazon AWS AI California, USA
Applied Scientist Intern Summer 2024

- Improve the safety of LLMs against jailbreaking attacks with steering

Academia Sinica*Research Assistant*

Taipei, Taiwan

2020 - 2021

- PI: Chi-Jen Lu
- Study which pre-finetuning tasks can improve pretrained language models [7]
- Compress large image generators and stabilize GANs training with knowledge distillation [10]

Amazon Alexa AI*Applied Scientist Intern*

California, USA

Spring 2020

- Hosts: Yang Liu and Dilek Hakkani-Tür
- Improve models' commonsense knowledge with specialized loss terms and learnable knowledge base [8, 9]

National Taiwan University*Research Assistant*

Taipei, Taiwan

2018 - 2020

- Advisor: Yun-Nung (Vivian) Chen
- Probing the degree of contextualization in ELMo & BERT embeddings with multisense word definitions [11]
- Develop LMs to automate diagnosis from clinical notes [12]

PUBLICATIONS

- [1] **Ting-Yun Chang**, Muru Zhang, Jesse Thomason, and Robin Jia. **Why Do Some Inputs Break Low-Bit LLM Quantization?** EMNLP 2025.
- [2] Ming Zhong, Xiang Zhou, **Ting-Yun Chang**, Qingze Wang, Nan Xu, Xiance Si, Dan Garrette, Shyam Upadhyay, Jeremiah Liu, Jiawei Han, Benoit Schillings, and Jiao Sun. **Vibe Checker: Aligning Code Evaluation with Human Preference**. arXiv 2025. Internship Project.
- [3] **Ting-Yun Chang**, Jesse Thomason, and Robin Jia. **When Parts Are Greater Than Sums: Individual LLM Components Can Outperform Full Models**. EMNLP 2024.
- [4] **Ting-Yun Chang**, Jesse Thomason, and Robin Jia. **Do Localization Methods Actually Localize Memorized Data in LLMs? A Tale of Two Benchmarks**. NAACL 2024.
- [5] **Ting-Yun Chang** and Robin Jia. **Data Curation Alone Can Stabilize In-context Learning**. ACL 2023.
- [6] Tejas Srinivasan, **Ting-Yun Chang**, Leticia Leonor Pinto Alva, Georgios Chochlakis, Mohammad Rostami, and Jesse Thomason. **CLiMB: A Continual Learning Benchmark for Vision-and-Language Tasks**. NeurIPS Datasets and Benchmarks Track 2022.
- [7] **Ting-Yun Chang** and Chi-Jen Lu. **Rethinking Why Intermediate-Task Fine-Tuning Works**. Findings of EMNLP 2021.
- [8] **Ting-Yun Chang**, Yang Liu, Karthik Gopalakrishnan, Behnam Hedayatnia, Pei Zhou, and Dilek Hakkani-Tür. **Go Beyond Plain Fine-tuning: Improving Pretrained Models for Social Commonsense**. IEEE SLT 2021.
- [9] **Ting-Yun Chang**, Yang Liu, Karthik Gopalakrishnan, Behnam Hedayatnia, Pei Zhou, and Dilek Hakkani-Tür. **Incorporating Commonsense Knowledge Graph in Pretrained Models for Social Commonsense Tasks**. DeeLIO Workshop at EMNLP 2020 (best paper award).
- [10] **Ting-Yun Chang** and Chi-Jen Lu. **TinyGAN: Distilling BigGAN for Conditional Image Generation**. Asian Conference on Computer Vision 2020.
- [11] **Ting-Yun Chang** and Yun-Nung Chen. **What Does This Word Mean? Explaining Contextualized Embeddings with Natural Language Definition**. EMNLP-IJCNLP 2019.

- [12] Shang-Chi Tsai, **Ting-Yun Chang**, and Yun-Nung Chen. [Leveraging Hierarchical Category Knowledge for Data-Imbalanced Multi-Label Diagnostic Text Understanding](#). LOUHI Workshop at EMNLP-IJCNLP 2019.
- [13] Chao-Chun Liang, Shih-Hong Tsai, **Ting-Yun Chang**, Yi-Chung Lin, and Keh-Yih Su. [A Meaning-based English Math Word Problem Solver with Understanding, Reasoning and Explanation](#). COLING 2016: System Demonstrations.

TEACHING EXPERIENCE

Teaching Assistant

USC CS 544 Applied Natural Language Processing, Fall 2024. Instructor: Swabha Swayamdipta.

USC CS 467 Introduction to Machine Learning, Spring 2023. Instructor: Robin Jia.

NTU CS Applied Deep Learning, Spring 2019. Instructor: Yun-Nung (Vivian) Chen.

PROGRAMMING

Languages: Python, C/C++, Java

Frameworks: PyTorch, TensorFlow, Triton, vLLM, verl